

Rainfall Frequency Analysis — SME Specification

Project: MPSB / Indargarh Hydrology Platform (pibuilder) **Prepared by:** pibuilder development team **Date:** April 2026 **Status:** Phase 1 complete — rainfall engine fully implemented and tested

1. What Was Built

A software engine that replicates and extends the rainfall frequency analysis currently performed manually in Excel. The engine automates the full workflow: from raw annual-maximum daily rainfall data through statistical distribution fitting, goodness-of-fit testing, and design rainfall selection.

Two data workflows are supported:

Workflow	Data source	Output
Grid-based (MPSB)	Satellite annual-max daily rainfall, 334 grid nodes, 0.25° spacing	100-year RP depth (mm) per node
Gauge-based (Indargarh)	32-year rain gauge record (1990–2021)	IDF table: mm/hr × 5 durations × 5 return periods

The engine is the first phase of a larger EPC hydrology platform that will eventually cover GIS watershed delineation, SCS/Rational Method discharge, Manning's HFL, and PDF report generation.

2. What the Engine Computes

For each data point the engine:

1. **Fits five probability distributions** to the annual-maximum series using the method of moments (matching Excel exactly):
2. Log-Pearson Type III (LP3) — the IS:5477 / CWC standard for Indian hydrology
3. Gumbel Extreme Value Type I (EV1)
4. Normal
5. Log-Normal (2-parameter)

6. Pearson Type III

7. **Tests each fitted distribution** using two independent goodness-of-fit tests:

8. **Chi-square test** — checks whether observed rainfall counts in fixed 50 mm bins ([50,100), [100,150), ..., [250,300) mm) match the theoretical distribution. Critical value: $\chi^2(0.95, df=2) = 5.991$.

9. **Kolmogorov-Smirnov test** — compares the empirical CDF against the theoretical CDF. Critical value: $1.36/\sqrt{n} = 0.304$ for $n = 20$ years.

10. **Selects the design distribution** as the one with the highest 100-year return-period rainfall among all distributions that pass both tests. This is the conservative choice — when multiple distributions are statistically acceptable, the larger estimate governs. If no distribution passes both tests, LP3 is used as a fallback (replicating Excel behaviour).

11. **Scales to sub-daily durations** (gauge workflow) using the CWC power-law:

$$P_t = P_{24} \times (t/24)^{(1/3)} \quad I_t = P_t / t \text{ (mm/hr)}$$

3. Validation

The engine was built using **Test-Driven Development against the Excel workbooks**:

- MPSB Rainfall Analysis_China Region.xlsx — 334 satellite grid points
- Indargarh Rainfall Analysis.xlsx — gauge station IDF table

Ground-truth values were extracted from Excel into JSON fixture files. The test suite runs entirely from these JSON files — no Excel dependency at test time.

Test results: 13 tests, all passing.

Test	Tolerance	Result
LP3 100-yr RP — all 334 MPSB grid points	0.01 mm	Pass
Gumbel 100-yr RP — all 334 MPSB grid points	0.01 mm	Pass
	0.5 mm	

Test	Tolerance	Result
Selected distribution — all 334 MPSB grid points		Pass (17 intentional divergences, see §4.2)
Indargarh IDF table — 25 cells (5 durations × 5 RPs)	0.001 mm/hr	Pass

4. Bugs Found in the Excel Workbook

During TDD, three discrepancies were identified between our implementation and the Excel workbook. Two are confirmed bugs in Excel; one was a bug in our initial implementation that has since been fixed.

4.1 Gumbel Sheet — STDEVP in Column 1 Only

Sheet: `gumbel (ks test)` — Row 24 ("Std Dev") — Affected point: (27.875°N, 89.125°E)

What it is: The first data column uses `STDEVP()` (population standard deviation, `ddof=0`). Every other column correctly uses `STDEV()` (sample standard deviation, `ddof=1`).

Evidence:

Column	Stored std in Excel	np.std ddof=0	np.std ddof=1
(27.875, 89.125) — Bug column	25.507	25.507 ✓	26.123 ✗
(28.125, 89.125) — Normal column	22.176	21.614 ✗	22.176 ✓

Impact: The 100-year RP at (27.875°N, 89.125°E) is 164.1 mm in Excel versus the correct value of 166.2 mm — a difference of ~2 mm. All other 333 Gumbel values are unaffected.

Our implementation: Uses sample standard deviation (`ddof=1`) everywhere.

Fix in Excel: Change `STDEVP()` to `STDEV()` in row 24, column B of the Gumbel sheet.

4.2 Chi-Square Sheets — FREQUENCY Formula Row Range Shift (Most Consequential)

Sheet: pearson(chi sq) and all other chi-sq sheets — Row 51 ("Obs Frequency") — Affected: 333/334 columns

What it is: The observed-frequency count uses Excel's FREQUENCY() function. In the first column the formula correctly references the 20 data rows:

```
Column B (first): =FREQUENCY(B11:B30, $C$2:$C$6)  ← rows 11–30 = 20 data years ✓
Column C onward: =FREQUENCY(C15:C34, $C$2:$C$6)  ← rows 15–34 = WRONG x
```

A copy-paste relative-reference error has shifted the range by +4 rows in every column after the first. Rows 15–34 contain only the **last 16 years** of data plus four **statistical summary values** (mean, std, Kn, Cs) that fall inside the rainfall bins and corrupt the observed frequency counts.

Evidence for point (29.125°N, 88.625°E):

Bin	Correct count (20 years)	Excel count (corrupted)	Cause of difference
[50, 100) mm	6	9	4 data years dropped; Kn=2.326 and Cs=1.41 counted here? No — they're < 50, lost entirely. Net: 3 extra counts from unknown redistribution
[100, 150) mm	12	9	Mean=117mm and Std=32mm landed here, but 4 years dropped
[150, 200) mm	1	1	Unchanged
[200, 250) mm	1	1	Unchanged

The mean (117 mm) and std (32 mm) fall in the [100,150) bin; Kn (2.326) and Cs (1.41) are below 50 mm and fall outside all bins, effectively dropping 4 data observations from the tally.

Impact: Inflated chi-square statistics cause distributions to fail the test at northern high-elevation points where they would otherwise pass. Selection changes at **17 grid points** — all at lat > 29°N where skewness Cs is large:

Point	Excel selects	Our engine selects	RP difference
(29.125, 88.625)	LP3 = 221.9 mm	Pearson = 222.7 mm	0.8 mm
(29.125, 90.625)	LogNorm = 245.3 mm	Pearson = 257.3 mm	12.0 mm
(29.125, 91.625)	LP3 = 200.4 mm	Pearson = 224.0 mm	23.6 mm
(29.125, 92.375)	LP3 = 203.0 mm	Pearson = 203.6 mm	0.6 mm
(29.375, 83.875)	LogNorm = 237.3 mm	Pearson = 241.9 mm	4.6 mm
(29.375, 85.625)	LP3 = 238.9 mm	Pearson = 206.3 mm	−32.6 mm
(29.375, 91.625)	LP3 = 225.4 mm	Gumbel = 227.2 mm	1.8 mm
(29.375, 91.875)	LP3 = 227.2 mm	Pearson = 231.5 mm	4.3 mm
(29.375, 94.375)	Gumbel = 197.5 mm	Normal = 174.5 mm	−23.0 mm
(29.875, 83.625)	LP3 = 295.9 mm	Pearson = 306.8 mm	10.9 mm
(29.875, 95.125)	LP3 = 397.8 mm	Pearson = 402.0 mm	4.2 mm
(30.125, 85.125)	LP3 = 270.9 mm	Pearson = 235.1 mm	−35.8 mm
(30.125, 94.625)	Gumbel = 159.6 mm	Pearson = 161.2 mm	1.6 mm
(30.375, 82.625)	LP3 = 295.1 mm	Pearson = 285.3 mm	−9.8 mm
(30.375, 91.375)	LogNorm = 267.5 mm	LP3 = 242.6 mm	−24.9 mm
(30.375, 95.625)	LP3 = 196.7 mm	Pearson = 198.8 mm	2.1 mm
(30.625, 82.625)	LP3 = 285.7 mm	Pearson = 291.2 mm	5.5 mm

Fix recommended in Excel: In row 51 of every chi-square sheet, the formula for **every column** must reference rows 11–30:

```
=FREQUENCY(<column>11:<column>30, $C$2:$C$6)
```

Our implementation: Always uses all 20 data values with the correct bins. The 17 discrepant points are explicitly listed in the test suite and emit a documented warning rather than a test failure.

4.3 Wilson-Hilferty Formula — Polynomial vs Exact (Our Bug, Now Fixed)

Note: Excel is correct here. This was a bug in our initial implementation.

What it is: The Wilson-Hilferty (1931) frequency factor K_T has two forms:

Exact (cube-root):

$$K_T = (2/Cs) \times [(1 - Cs^2/36 + z \cdot Cs/6)^3 - 1]$$

Polynomial (5-term Taylor expansion, common in textbooks):

$$K_T \approx z + (z^2-1)k + (z^3-6z)k^2/3 - (z^2-1)k^3 + zk^4 + k^5/3, \quad k = Cs/6$$

Both are equivalent for small $|Cs|$. For $|Cs| > \sim 1.9$ the polynomial diverges.

Evidence at (29.375°N, 94.375°E), $Cs = -2.103$:

Method	K_T value	Error vs exact
Exact cube-root (Excel & our engine)	0.95084110	—
Polynomial (initial implementation)	0.94731537	0.00353
scipy pearson3 (different parameterisation)	0.94484135	0.00600

The polynomial error of 0.0035 in K_T propagates to ~ 0.17 mm in the final RP — above the 0.01 mm test tolerance.

Our implementation: Fixed. Uses the exact cube-root formula. Matches Excel to 8 decimal places across all 334 grid points.

5. Bug Summary and Recommendations

#	Location	Scope	Max RP impact	Action required in Excel
1	Gumbel sheet, column 1	1 point	~ 2 mm	Change <code>STDEVP</code> → <code>STDEV</code> in row 24, col B
2	All chi-sq sheets, cols 2-334	333 points; 17 selections change	~ 36 mm	Fix <code>FREQUENCY</code> range to rows 11-30 in all columns

#	Location	Scope	Max RP impact	Action required in Excel
3	WH polynomial (our bug — fixed)	Would affect 5 points	~0.3 mm	N/A — resolved in engine

Bug 2 is the most consequential. At 17 northern grid points (lat > 29°N) our engine selects a different distribution than Excel because the correct chi-square test passes Pearson III where Excel's corrupted test rejects it. Whether Pearson III is the better engineering choice at those points is a question for SME review.

6. Query Tool

A command-line tool is available to look up the 100-year design rainfall at any coordinate within the MPSB grid coverage:

Grid coverage: 27.875°–31.125°N, 82.125°–95.625°E

```
.venv/bin/python scripts/rainfall_query.py <lat> <lon>
```

Example — query at (29.0°N, 88.5°E):

```
Query:          (29.0, 88.5)
Nearest grid: (28.875, 88.375) [19.6 km away]
Data years:    20 (annual-max daily rainfall, mm)
```

100-year design rainfall:	177.8 mm
Selected distribution:	Gumbel

Distribution	100yr RP (mm)	Chi-sq	KS	Selected
LP3	172.59	PASS	PASS	
Gumbel	177.78	PASS	PASS	<-- SELECTED
Normal	158.21	PASS	PASS	
Log Normal	170.45	PASS	PASS	
Pearson	173.11	PASS	PASS	

7. Pending Phases

Component	Description
GIS engine	DEM-based watershed delineation (WhiteboxTools), basin area/slope
Hydrology engine	SCS Curve Number runoff depth, Rational Method + SCS peak discharge
Hydraulics engine	Manning's equation HFL, afflux, freeboard (IRC:5-2015)
Data pipeline	DEM, shapefile, and rainfall raster ingestion and ETL
API layer	FastAPI service layer
Report generation	Jinja2/WeasyPrint automated PDF reports

8. Key References

Reference	Used for
Wilson & Hilferty (1931). "The distribution of chi-square." PNAS 17(12).	Exact K _T formula
Gumbel, E.J. (1958). Statistics of Extremes. Columbia University Press.	Gumbel EV1
Chow, Maidment & Mays (1988). Applied Hydrology. McGraw-Hill.	General frequency analysis
IS:5477 Part 4 (1971). Bureau of Indian Standards.	LP3 as Indian hydrology standard
CWC (2002). Flood Estimation — Western Himalayas Zone 7.	Duration scaling exponent 1/3
Kolmogorov (1933); Smirnov (1948).	KS test critical values